Exercise III

1. If G_1, G_2 are the geometric means of two series of observations and G is the geometric mean of the ratios of the corresponding observations, then the value of G : **[AIEEE-2002]**

प्रेक्षणों की दो श्रेणियों के गुणोत्तर माध्य क्रमशः G_1, G_2 हैं तथा संगत प्रेक्षणों के अनुपातों का गुणोत्तर माध्य G हो तो G का मान होगा : **[AIEEE-2002]**

(A) $\frac{G_1}{G_2}$ (B) $\frac{\log G_1}{\log G_2}$ (C) $\log (G_1 . G_2)$ (D) $\log G_1 - \log G_2$

Ans. A

Sol. Let $x_1, x_2, ..., x_n$ be n observation corresponding to variate x and $y_1, y_2, ... y_n$ be n observation corresponding to variate y

$$G_1 = \left(x_1.x_2....x_n\right)^{\frac{1}{n}}$$

$$G_2 = \left(y_1.y_2....y_n\right)^{\frac{1}{n}}$$

Now the geometric mean of $\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, ... \frac{x_n}{y_n}$ is

$$G = \left(\frac{x_1}{y_1}.\frac{x_2}{y_2}.\frac{x_3}{y_3}.....\frac{x_n}{y_n}\right)^{\frac{1}{n}}$$

$$= \frac{\left(x_1.x_2.x_3.....y_n\right)^{\frac{1}{n}}}{\left(y_1.y_2.y_3.....y_n\right)^{\frac{1}{n}}}$$

$$G = \frac{G_1}{G_2}$$

2. The median of a set of 9 distinct observations is 20.5 If each of the largest 4 observations of the set is increased by 2, then the median of the new set : **[AIEEE-2003]**

(A) remains the same as that of the original set (B) is increased by 2

(C) is decreased by 2 (D) is two times the original median

9 अलग अलग प्रेक्षणों की माध्यिका 20.5 है। यदि महत्तम 4 प्रेक्षणों में से प्रत्येक में 2 की वृद्धि कर दी जाए, तो नये समुच्चयों की माध्यिका होगी : **[AIEEE-2003]**

(A) समान रहेगी (B) 2 की वृद्धि होगी (C) 2 का ह्रास होगा (D) दुगुनी हो जाऐगी

Ans. A

Sol. Given the number of observation is 9. Therefore median is $\left(\frac{n+1}{2}\right)^{th}$ value $= \left(\frac{9+1}{2}\right)^{th} = 5^{th}$ term

The value of the largest four observations is increased by 2 then middle term remain unchanged. So the median of new observations is also the 5th value. Which is not changed.

3. In an experiment with 15 observations on x, the following results were available

$\sum x^2 = 2830$, $\sum x = 170$

One observation that was 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is : **[AIEEE-2003]**

एक प्रयोग में x के 15 प्रेक्षण हैं, जिसके परिणाम निम्न हैं :

$$\sum x^2 = 2830,\ \sum x = 170$$

एक प्रेक्षण जिसका मान 20 था, गलत था तथा उसको सही मान 30 से प्रतिस्थापित किया जाता है, तो सही प्रसरण होगा : **[AIEEE-2003]**

(A) 8.33 (B) 78.00 (C) 188.66 (D) 177.33

Ans. B

Sol. $\sum x^2 = 2830$

$\sum x = 170$

$n = 15$

Given that one observation 20 was replaced by 30

$\therefore \sum x = 170 - 20 + 30 = 180$

$\sum x^2 = 2830 - (20)^2 + (30)^2 = 3330$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

$$= \frac{3330}{15} - \left(\frac{180}{15}\right)^2 = 78$$

4. Consider the following statements : **[AIEEE-2004]**

(a) Mode can be computed from histogram

(b) Median is not independent of change of scale

(c) Variance is independent of change of origin and scale.

Which of these is/ are correct ?

(A) only (a) (B) only (b) (C) only (a) and (b) (D) (a), (b) and (c)

निम्न कथनों को देखें : **[AIEEE-2004]**

(a) बहुलक आयत चित्रा से परिकलित किया जा सकता है

(b) माध्यक मापन के स्केल से स्वतंत्रा नहीं है

(c) प्रसरण मूल बिन्दु तथा मापन के स्केल से स्वतंत्रा है

(d) उपरोक्त में से कौनसा सत्य है ?

(A) केवल (a) (B) केवल (b) (C) केवल (a) तथा (b) (D) (a), (b) तथा (c)

Ans. C

Sol.

5. In a series of 2n observations, half of them equal a and remaining half equal – a. If the standard deviation of the observations is 2, then |a| equals : **[AIEEE-2004]**

2n क्रमित प्रेक्षणों में से आधे a के बराबर हैं तथा शेष आधे – a के बराबर है। यदि प्रेक्षणों का मानक विचलन 2 है, तो |a| बराबर है : **[AIEEE-2004]**

(A) 1/n (B) $\sqrt{2}$ (C) 2 (D) $\frac{\sqrt{2}}{n}$

Ans. C

Sol. Total number of observations = 2n

Half are equal to a and half are equal to –a therefore mean $\overline{x} = 0$

Standard deviation $\sigma = \sqrt{\frac{\left|x_i - \overline{x}\right|^2}{2n}}$

$$2 = \sqrt{\frac{2na^2}{2n}}$$

$2 = |a|$

6. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately : **[AIEEE-2005]**

यदि किसी बारंबारता बंटन का माध्य तथा माध्यिका क्रमशः 21 तथा 22 है, तो उसका अनुमानित बहुलक है : **[AIEEE-2005]**

(A) 22.0 (B) 20.5 (C) 25.5 (D) 24.0

Ans. D

Sol. mean = 21

medain = 22

mode = 3 median – 2 mean

= 3.22 – 2.21

= 66 – 42

= 24

7. Let $x_1, x_2, ... x_n$ be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is: **[AIEEE-2005]**

माना $x_1, x_2, ... x_n$ ऐसे n प्रेक्षण हैं, कि $\sum x_i^2 = 400$ तथा $\sum x_i = 80$ तो निम्न में से n का एक संभावित मान है: **[AIEEE-2005]**

(A) 15 (B) 18 (C) 9 (D) 12

Ans. B

Sol. $\sum x_i^2 = 400$

$\sum x_i = 80$

RMS $\geq$ AM

$$\sqrt{\frac{\sum x_i^2}{n}} = \frac{\sum x_i}{n}$$

$$\sqrt{\frac{400}{n}} \geq \frac{80}{n}$$

$$\frac{400}{n} \geq \frac{6400}{n^2} \quad \Rightarrow \frac{1}{n} \geq \frac{16}{n^2}$$

$$\Rightarrow \frac{n-16}{n^2} \geq 0 \qquad \Rightarrow n \geq 16$$

8. Suppose a population A has 100 observations 101, 102,.... 200, and another population B has 100 observations 151, 152,.....250. If V_A and V_B represent the variances of the two populations, respectively, then $\frac{V_A}{V_B}$ is: **[AIEEE-2006]**

माना समष्टि A में 101, 102,.... 200 के 100 प्रेक्षण है तथा एक अन्य समष्टि B में 151, 152,.....250 के 100 अन्य प्रेक्षण है। यदि इन दो समष्टियों के प्रसरण क्रमशः V_A तथा V_B है, तो $\frac{V_A}{V_B}$ का मान है : **[AIEEE-2006]**

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$ (C) 2 (D) 1

Ans. D

Sol. $\sigma_x^2 = \frac{\sum d_i^2}{n}$

Here the deviations are taken from the mean. since A and B have both 100 conseultive integers, therefore both have same standard deviations and varience.

$\therefore \frac{V_A}{V_B} = 1$

9. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is : **[AIEEE-2007]**

एक कक्षा के छात्रों के अंको का औसत 52 तथा छात्राओं के अंको का औसत 42 है। छात्रों तथा छात्राओं के संयुक्त अंको का औसत 50 है। कक्षा में छात्रों का प्रतिशत होगा : **[AIEEE-2007]**

(A) 40 (B) 20 (C) 80 (D) 60

Ans. C

Sol. Let the number of boys and girls be x and y

$\therefore 52x + 42y = 50(x + y)$

$52x + 42y = 50x + 50y$

$52x - 50x = 50y - 42y$

$2x = 8y$

$x = 4y$

total number of students in the class $x + y = 5y$

Percentage of boys $\frac{x}{x+y} \times 100$

$= \frac{4y}{4y+y} \times 100 \qquad = 80\%$

10. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b ? **[AIEEE-2008]**

संख्याओं a, b, 8, 5, 10 का माध्य 6 है तथा प्रसरण 6.80 है। तो निम्न में से कौनसा एक a तथा b के सम्भावित मान देगा? **[AIEEE-2008]**

(A) a = 5, b = 2 (B) a = 1, b = 6 (C) a = 3, b = 4 (D) a = 0, b = 7

Ans. C

Sol. Given numbers = a, b, 8, 5, 10

mean = 6

$$\frac{a+b+8+5+10}{5}=6$$

$a + b = 7$

$b = 7 - a$

varience = 6.8

$$\frac{(a-6)^2+(b-6)^2+(8-6)^2+(5-6)^2+(10-6)^2}{5}=6.8$$

$$a^2+b^2-12a-12b+36+36+4+1+16=34$$

$$a^2+b^2-129-12b=-59 \qquad \text{_____(2)}$$

$$a^2+(7-9)^2-129-12(7-9)=59$$

$$2a^2-14a+24=0$$

$$a^2-7a+12=0$$

$$(a-4)(a-3)=0$$

a = 4 or 3

b = 3 or 4

11. If the mean deviation of the numbers 1, 1 + d, 1+ 2d, …..1 +100d from their mean is 255, then the d is equal to : **[AIEEE-2009]**

यदि संख्याओं 1, 1 + d, 1+ 2d, …..1 +100d का उनके माध्य से माध्य विचलन 255 है, तब d बराबर है : **[AIEEE-2009]**

(A) 10.0 (B) 20.0 (C) 10.1 (D) 20.2

Ans. C

Sol. Given observations = 1, 1 + d, 1 + 2d, ..., 1+ 100d

$$\text{sum of observations} = \frac{101}{2}[1+1+100d]$$

$$= \frac{101}{2}[2+100d]$$

$$= 101(1+50d)$$

$$\text{mean} = \frac{101(1+50d)}{101} = 1+50d$$

$$\text{mean deviation} = \frac{\sum|x-x_i|}{n}$$

$$= \frac{|1+50d-1|+|1+50d-1-d|+.....+|1+100d-1-50d|}{101}$$

$$= \frac{50d+49d+.....+0+d+2d+....+50d}{101}$$

$$= \frac{2d(1+2+.....+50)}{101}$$

$$= \frac{2d(50)(51)}{101\times 2}$$

$$= \frac{50\times 51\times d}{101}$$

$$\text{mean deviation} = \frac{50\times 51\times d}{101} = 255$$

$$d = \frac{101\times 255}{50\times 51}$$

$$d = 10.1$$

12. **Statement – 1 :** The variance of first *n* even natural numbers is $\frac{n^2-1}{4}$

Statement – 2 : The sum of first *n* natural numbers is $\frac{n(n+1)}{2}$ and the sum of squares of first *n* natural numbers is $\frac{n(n+1)(2n+1)}{6}$: **[AIEEE-2009]**

(A) Statement -1 is true, Statement -2 is true; Statement-2 is a correct explanation for Statement -1

(B) Statement -1 is true, Statement -2 is true; Statement -2 is ***not*** a correct explanation for Statement -1.

(C) Statement -1 is true, Statement -2 is false. (D) Statement -1 is false, Statement -2 is ture.

कथन– **1 :** प्रथम *n* सम धन पूर्णाकों का प्रसरण $\frac{n^2-1}{4}$ है।

कथन– **2** : प्रथम *n* धन पूर्णाकों का योगफल $\frac{n(n+1)}{2}$ और प्रथम *n* धन पूर्णाकों के वर्गों का योगफल $\frac{n(n+1)(2n+1)}{6}$ है।

[AIEEE-2009]

(A) कथन-1 सही और कथन-2 सही है, कथन-2, कथन-1 का सही स्पष्टीकरण है

(B) कथन-1 सही है, कथन-2 सही है, कथन-2, कथन-1 का सही स्पष्टीकरण नहीं है

(C) कथन-1 सही है, कथन-2 गलत है

(D) कथन-1 गलत है, कथन-2 सही है

Ans. D

Sol. Sum of first n even natural numbers

$= 2 + 4 + 6 + ... + 2n$

$= 2(1 + 2 + ... + n)$

$= n(n + 1)$

$\text{mean}(\bar{x}) = \frac{n(n+1)}{n}$

$\bar{x} = n + 1$

$\text{Varience} = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$

$= \frac{1}{n}\left\{\left[2^2 + 4^2 + ... + \left(2n^2\right)\right]\right\} - (n+1)^2$

$= \frac{1}{n}\left\{2^2\left\{1^2 + 2^2 + ... + n^2\right\}\right\} - (n+1)^2$

$$= \frac{1}{n}\left[\frac{4n(n+1)(2n+1)}{6}\right] - (n+1)^2$$

$$= \frac{2(n+1)(2n+1)}{3} - (n+1)^2$$

$$= \left(\frac{n+1}{3}\right)(n-1)$$

$$= \frac{n^2-1}{3}$$

So statement-I is wrong but statement-II is correct

13. For two data sets each of size is 5, the variances are given to be 4 and 5 and the corresponding measn are given to be 2 and 4 respectively, then the variance of the combined data set is : **[AIEEE-2010]**

आकड़ों के दो समुच्चयों के लिए, जिनमें प्रत्येक में 5 अवयव है, के प्रसरण क्रमशः 4 तथा 5 है तथा उनके संगत माध्य क्रमशः 2 तथा 4 है। उनके संयुक्त आंकड़ों के समुच्चय का प्रसरण है : **[AIEEE-2010]**

(A) $\frac{5}{2}$ (B) $\frac{11}{2}$ (C) 6 (D) $\frac{13}{2}$

Ans. B

Sol. $\sigma_x^2 = 4, \quad \overline{x} = 2$

$\sigma_y^2 = 5, \quad \overline{y} = 4$

$$\sigma_x^2 = \frac{\Sigma x_i^2}{5} - (\overline{x})^2 = 4$$

$$\Sigma x_i^2 = 40$$

$$\sigma_y^2 = \frac{\sum y_i^2}{5} - (\overline{y})^2 = 5$$

$$\sum y_i^2 = 105$$

$$\sum (x_i^2 + y_i^2) = 40 + 105 = 145$$

Variance of combined data

$$= \frac{\left(\sum x_i^2 + \sum y_i^2\right)}{10} - \left(\frac{\sum x_i + \sum y_i}{10}\right)^2$$

$$\frac{145}{10} - \left(\frac{10+20}{10}\right)^2$$

$$= \frac{145}{10} - 9$$

$$\Rightarrow \frac{11}{2}$$

14. If the mean deviation about the median of the number a, 2a, ..., 50a is 50, then |a| equal : **[AEEE-2011]**

संख्याओं a, 2a, ..., 50a का माध्यक के सापेक्ष माध्य विचलन 50 है, तो |a| बराबर है : **[AEEE-2011]**

(A) 4 (B) 5 (C) 2 (D) 3

Ans. A

Sol. Observations : a, 2a, ... 50a

median is the mean of 25th and 26th observation

$$M = \frac{25a + 26a}{2} = 25.5a$$

mean deviation about median

$$M.D.(M) = \frac{\sum(x_i - M)}{N}$$

$$50 = \frac{1}{50}\left[2 \times |a| \times (0.5 + 1.5 + 2.5 + ... + 24.5)\right]$$

$2500 = |a| \times 25 \times 25$

$|a| = 4$

15. A scientist is weighing each of 30 fishes. Their mean weight worked out is 30 gm and a standard deviation of 2 gm. Later, it was found that the measuring scale was misaligned and always under reported every fish weight by 2 gm. The correct mean and standard deviation (in gm) of fishes are respectively : **[AIEEE-2011]**

एक वैज्ञानिक 30 मछलियों में से प्रत्येक को तोलता है। उनके भार का माध्य 30 ग्राम है तथा मानक विचलन 2 ग्राम है। बाद में यह पाया गया कि तोलने की मशीन, ठीक न होने के कारण प्रत्येक मछली का भार 2 ग्राम कम बताती है, तो मछलियों का सही माध्य तथा मानक विचलन (gm में) है। **[AIEEE-2011]**

(A) 28, 4 (B) 32, 2 (C) 32, 4 (D) 28, 2

Ans. B

Sol. Given that mean weight of fishes is 30gm

Standard deviation is 2gm

Given that every fish weight is increased by 2gm

Therefore overall mean weight = 2130

= 32gm

Standard deviation is not effected by the addition of a constant. Therefore standard deviation remains same.

16. Let $x_1, x_2, ..., x_n$ be n observations, and let $\bar{x}$ be their arithmetic mean and σ^2 be their variance.

Statement -1: Variance of $2x_1, 2x_2, ... 2x_n$ is $4\sigma^2$.

Statement-2: Arithmetic mean of $2x_1, 2x_2, ... 2x_n$ is $4\bar{x}$. **[AIEEE-2012]**

(A) Statement-1 is true, Statement-2 is false. (B) Statement-2 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(D) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

माना n प्रेक्षण $x_1, x_2, ..., x_n$ हैं तथा उनका गणितीय माध्य $\bar{x}$ तथा प्रसरण σ^2 है।

कथन-1: $2x_1, 2x_2, ... 2x_n$ का प्रसरण $4\sigma^2$ है।

कथन-2: $2x_1, 2x_2, ... 2x_n$ का गणितीय माध्य $4\bar{x}$ है। **[AIEEE-2012]**

(A) कथन-1 सत्य है, कथन-2 असत्य है।

(B) कथन-2 असत्य है, कथन-2 सत्य है।

(C) कथन-1 सही है, कथन-2 सही है; कथन-2, कथन-1 की सही व्याख्या है।

(D) कथन-1 सही है, कथन-2 सही है; कथन-2, कथन-1 की सही व्याख्या नहीं है।

Ans. A

Sol. $\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$

Varience of $2x_1, 2x_2, ...2x_n$

$= \frac{\sum (2x_i)^2}{n} - \left(\frac{\sum 2x_i}{n}\right)^2$

$$= 4\frac{\sum x_i^2}{n} - 4\left(\frac{\sum x_i}{n}\right)^2$$

$$= 4\left\{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2\right\}$$

$= 4\sigma^2$

So stalement 1 is true

mean of $x_1, x_2, ...x_n$

$$\overline{x} = \frac{x_1 + x_2 + ... + x_n}{n}$$

mean of $2x_1, 2x_2, ...2x_n$

$$= \frac{2x_1 + 2x_2 + ... + 2x_n}{n}$$

$$= 2\left(\frac{x_1 + x_2 + ... + x_n}{n}\right)$$

$= 2\overline{x}$

Statement 2 is incorrect

17. All the students of a class performed poorly in Mathemeatics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given ? **[JEE(Main)-2013]**

(A) mean (B) median (C) mode (D) variance

एक कक्षा के सभी विद्यार्थियों ने गणित में कम अंक प्राप्त किए। अध्यापक ने प्रत्येक विद्यार्थी को 10 रियायती अंक देने का निर्णय लिया। रियायती अंक देने के बाद भी निम्न में से कौनसा सांख्यिकी मान नहीं बदलेगा? **[JEE(Main)2013]**

(A) माध्य (B) माध्यक (C) बहुलक (D) प्रसरण

Ans. D

Sol. The varience and standard deviation doesn't change even after addition of some constant to the given observations.

18. The variance of first 50 even natural numbers is : **[JEE(Main)2014]**

पहली 50 सम प्राकृत संख्याओं का प्रसरण है : **[JEE(Main)2014]**

(A) 437 (B) $\frac{437}{4}$ (C) $\frac{833}{4}$ (D) 833

Ans. D

Sol. $25\overline{x} = \frac{2+4+6+....+100}{50} \times 25$

$\overline{x} = 51$

$\sigma^2 = \frac{\sum x_i^2}{n} - (\overline{x})^2$

$\sigma^2 = \frac{4(1^2+2^2+.....+50^2)}{50} - (51)^2$

$= 833$

19. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3, 4 and 5 and added to the data, then the mean of the resultant data, is :

[JEE(Main)-2015]

16 प्रेक्षणों वाले आँकड़ों का माध्य 16 है। यदि एक प्रेक्षण जिसका मान 16 है, को हटा कर 3 नये प्रेक्षण जिनके मान 3, 4 तथा 5 हैं, आँकड़ों मे मिला दिये जाते हैं, तो नये आँकड़ों का माध्य है : **[JEE(Main)2015]**

(A) 16.8 (B) 16.0 (C) 15.8 (D) 14.0

Ans. D

Sol. There are 16 observations and the mean is 16.

Hence the initial sum = 16 × 16

= 256

after adding the new observations and deleting observation of 16, then the sum is = 256 – 16 + 3 + 4 + 5 = 252

Number of observations = 16 – 1 + 3

= 18

New average $= \frac{252}{18}$

= 14

20. If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ?

[JEE(Main)-2016]

यदि संख्याओं 2, 3, a तथा 11 का मानक विचलन 3.5 है, तो निम्न में से कौन–सा सत्य है? **[JEE(Main)2016]**

(A) $3a^2 - 34a + 91 = 0$ (B) $3a^2 - 23a + 44 = 0$

(C) $3a^2 - 26a + 55 = 0$ (D) $3a^2 - 32a + 84 = 0$

Ans. D

Sol. $\text{Varience} = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$

$$(3.5)^2 = \frac{4+9+a^2+121}{4} - \left(\frac{16+a}{4}\right)^2$$

$$3a^2 - 32a + 280 = 16 \times \frac{49}{4} = 4 \times 49$$

$$3a^2 - 32a + 84 = 0$$

21. If $\sum_{i=1}^{9}(x_i - 5) = 9$ and $\sum_{i=1}^{9}(x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, ..., x_9$ is :

[JEE(Main)-2018]

यदि $\sum_{i=1}^{9}(x_i - 5) = 9$ तथा $\sum_{i=1}^{9}(x_i - 5)^2 = 45$ है, तो नौ प्रेक्षणों $x_1, x_2, ..., x_9$ का मानक विचलन है :

[JEE(Main)2018]

(A) 2 (B) 3 (C) 9 (D) 4

Ans. A

Sol. $\sum(x_i - 5) = 9$

$i = 1$

$$\sum_{i=1}^{n}(x_i - 5)^2 = 45$$

$$\text{Standard deviation} = \sqrt{\sum_{i=1}^{9}\left[(x_i - 5)\right]^2 - \left[\frac{\sum(x_i - 5)}{9}\right]^2}$$

$$= \sqrt{\frac{45}{9} - \left(\frac{9}{9}\right)^2}$$

$$= \sqrt{5-1}$$

$$= 2$$

22. The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to : **[JEE(Main)Jan-2020]**

10 प्रेक्षणों के माध्य तथा मानक विलचन क्रमशः 20 तथा 2 हैं। इन 10 प्रेक्षणों में से प्रत्येक को p से गुणा करने के पश्चात् प्रत्येक में से q कम किया गया, जहां $p \neq 0$ तथा $q \neq 0$ हैं। यदि नए माध्य तथा मानक विचलन के मान अपने मूल मानों के आधे हैं, तो q का मान है : **[JEE(Main)Jan.2020]**

(A) –20 (B) –10 (C) –5 (D) 10

Ans. A

Sol. New mean $\bar{x}_1 = p\bar{x} - q$

$\therefore \bar{x}_1 = \frac{\bar{x}}{2} = 10$

$p\bar{x} - q = 10$

$20p - q = 10$ _____(1)

new standard deviation $\sigma_1 = |p|\sigma$

$\sigma_1 = \frac{\sigma}{2} = 1$

$\Rightarrow 2|p| = 1 \Rightarrow p = \pm\frac{1}{2}$

If $p = -\frac{1}{2}$ $\Rightarrow q = -20$

23. The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is : **[JEE(Main)Jan-2020]**

20 प्रेक्षणों के माध्य तथा प्रसरण क्रमशः 10 तथा 4 पाये गये। पुनः जांच करने पर पाया गया कि एक प्रेक्षण 9 गलत था तथा सही प्रेक्षण 11 था। तो सही प्रसरण है : **[JEE(Main)Jan.2020]**

(A) 3.98 (B) 4.01 (C) 3.99 (D) 4.02

Ans. C

Sol. $\bar{x} = 10 \Rightarrow \sum x_i = 20 \times 10 = 200$

$\sigma^2 = 4 \Rightarrow \frac{\sum x_i^2}{20} - (\bar{x})^2 = 4$

$\sum x_i^2 = 2080$

Actual $\sum x_i = 200 - 9 + 11 = 202$

$\sum x_i^2 = 2080 - 9^2 + 11^2 = 2120$

$$\sigma_1^2 = \frac{2120}{20} - \left(\frac{202}{20}\right)^2$$

$$\sigma_1^2 = 3.99$$

24. Let the observations $x_i\,(1 \le i \le 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$. If μ and λ are the mean and the variance of the observations, $x_1 - 3, x_2 - 3, ..., x_{10} - 3$, then the ordered pair (μ, λ) is equal to : **[JEE(Main)Jan-2020]**

माना प्रेक्षण $x_i\,(1 \le i \le 10)$ समीकरणों $\sum_{i=1}^{10}(x_i - 5) = 10$ तथा $\sum_{i=1}^{10}(x_i - 5)^2 = 40$ को संतुष्ट करते हैं। यदि μ तथा λ प्रेक्षणों $x_1 - 3, x_2 - 3, ..., x_{10} - 3$ के क्रमशः माध्य तथा प्रसरण हैं, तो क्रमित युग्म (μ, λ) बराबर है : **[JEE(Main)Jan.2020]**

(A) (6, 3) (B) (6, 6) (C) (3, 6) (D) (3, 3)

Ans. D

Sol. $$\sum_{i=1}^{10}(x_i - 5) = 10 \quad .....(1)$$

$$\sum_{i=1}^{10}(x_i - 5)^2 = 40 \quad .....(2)$$

$$\text{mean}\,(x_i - 5) = \frac{\sum(x_i - 5)}{10} = \frac{10}{10} = 1$$

$$\mu = \text{mean}\,(x_i - 3) = \text{mean}\,(x_i - 5 + 2) = 1 + 2 = 3$$

$$\lambda = \text{var}(x_i - 3) = \text{var}(x_i - 5) = \frac{\sum(x_i - 5)^2}{10} - \frac{\sum(x_i - 5)}{10}$$

$$= \frac{40}{10} - 1$$

$$= 4 - 1 = 3$$

25. If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then m + n is equal to _____. **[JEE(Main)Jan-2020]**

यदि प्रथम n प्राकृत संख्याओं का प्रसरण 10 है और प्रथम m सम–प्राकृत संख्याओं का प्रसरण 16 है, तो m + n बराबर है _____ **[JEE(Main)Jan.2020]**

Ans. 18

Sol. Variance of first n-natural numbers = 10

$\Rightarrow \frac{\sum n^2}{n} - \left(\frac{\sum n}{n}\right)^2 = 10$ $\Rightarrow \frac{n(n+1)(2n+1)}{6\times n} - \left(\frac{n(n+1)}{2n}\right)^2 = 10$

$\Rightarrow \frac{2(2n^2+3n+1)-3(n^2+2n+1)}{12} = 10$

$\Rightarrow n^2 - 1 = 120$ $\Rightarrow n^2 = 121$ $\Rightarrow n = \pm 11$

But since $n \in N \Rightarrow x = 11$

Similarly variance of first m even natural numbers = 16

$\Rightarrow$ Variance of first m natural numbers = 4

$\Rightarrow \frac{m^2-1}{12} = 4$ $\Rightarrow m^2 = 49$ $\Rightarrow m = 7$

$m + n = 7 + 11 = 18$

26. If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then $x \cdot y$ is equal to _____ . **[JEE(Main)Jan-2020]**

यदि आठ संख्याओं 3, 7, 9, 12, 13, 20, x तथा y के माध्य तथा प्रसरण क्रमशः 10 तथा 25 हैं, तो $x \cdot y$ बराबर है _____ **[JEE(Main)Jan.2020]**

Ans. 54

Sol. Mean = $\frac{\sum x_i}{8} = 10$

$x + y + 64 = 80$

$x + y = 16$ .....(1)

$\sigma^2 = \frac{\sum x_i^2}{8} - \left(\frac{\sum x_i}{8}\right)^2 = 25$

$\frac{x^2+y^2+852}{8} - 100 = 25$

$\frac{x^2+y^2+852}{8} = 125$

$x^2 + y^2 = 148$ .....(2)]

$(x + y)^2 = x^2 + y^2 + 2xy$

$xy = 54$

27. Let $X = \{x \in N : 1 \le x \le 17\}$ and $Y = \{ax + b : x \in X \text{ and } a, b \in R, a > 0\}$. If mean and variance of elements of Y are 17 and 216 respectively then a + b is equal to : **[JEE(Main)Sept-2020]**

माना $X = \{x \in N : 1 \le x \le 17\}$ तथा $Y = \{ax + b : x \in X$ तथा $a, b \in R, a > 0\}$ हैं। यदि Y के अवयवों के माध्य तथा प्रसरण क्रमशः 17 तथा 216 हैं, तो $a + b$ बराबर है : **[JEE(Main)Sept.2020]**

(A) 9 (B) – 7 (C) 7 (D) – 27

Ans. B

Sol. $X = \{1, 2, ...17\}$

$$\text{Mean} = \frac{1+2+.....+17}{17} = 9$$

$$\text{Variance} = \frac{\sum x_1^2}{17} - \left(\bar{x}\right)^2 = 105 - 81 = 24$$

$Y = \{ax + b : x \in X$ and $a, b \in R, a > 0\}$

Mean = $9a + b = 17$

Variance = $24a^2 = 216$

$\Rightarrow a^2 = 9 \Rightarrow a = 3$

$b = -10$

28. If the variance of the terms in an increasing A.P., $b_1, b_2, b_3, ..., b_{11}$ is 90, then the common difference of this A.P. is _____. **[JEE(Main)Sept-2020]**

यदि $b_1, b_2, b_3, ..., b_{11}$ एक वर्धमान A.P. है और इसके पदों का प्रसरण 90 है, तो इस A.P. का सार्व अन्तर है_____ **[JEE(Main)Sept.2020]**

Ans. 3

Sol. $b_1, b_2, ..., b_{11}$ are in A.P.

Let a and d be the first term and common difference respectively.

Variance $(b_1, b_2, ..., b_{11}) = 90$ ..... (1)

$$\text{Mean} = \frac{\frac{11}{2}\left[2a + (11-1)d\right]}{11} = a + 5d = \bar{b}$$

$$\text{Variance} = \frac{1}{11}\sum_{i=1}^{11}\left(b_i - \bar{b}\right)^2$$

$$= \frac{1}{11}\sum_{i=1}^{11}\left(a + (i-1)d - (a+5d)\right)^2$$

$$= \frac{1}{11}\sum_{i=1}^{11}\left((i-6)d\right)^2$$

$$= \frac{d^2}{11}\left[5^2+4^2+3^2+2^2+1^2+0+1^2+2^2+....+5^2\right]$$

$$= \frac{2d^2}{11}\left[1^2+2^2+....+5^2\right]$$

$$= \frac{2d^2}{11}\left[\frac{5(6)11}{6}\right] = 10d^2$$

We have

$\because 10d^2 = 90$

$d^2 = 9$

$d = 3$ [as A.P. is an increasing one]

29. For the frequency distribution :

Variate (x) :	x_1	x_2	x_3 ..... x_{15}
Frequency (f) :	f_1	f_2	f_3 ..... f_{15}

where $0 < x_1 < x_2 < x_3 < ... < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, the standard deviation cannot be :

[JEE(Main)Sept-2020]

बारंबारता बंटन

चर (x) :	x_1	x_2	x_3 ..... x_{15}
बारंबारता (f) :	f_1	f_2	f_3 ..... f_{15}

जहाँ $0 < x_1 < x_2 < x_3 < ... < x_{15} = 10$ तथा $\sum_{i=1}^{15} f_i > 0$ है, का मानक विचलन, निम्न में से कौन–सा नहीं हो सकता है :

[JEE(Main)Sept.2020]

(A) 6 (B) 1 (C) 2 (D) 4

Ans. A

Sol. standard deviation will be maximum when half x_i approch to 0 are half to 10.

$\bar{x}$ will be middle value = 5

$$\left(x_i - \bar{x}\right)^2 \le 25$$

$$\left(x_i - \bar{x}\right) \le 5 \; ; \; \sigma \le 5$$

30. Let $x_i (1 \le i \le 10)$ be ten observation of a random variable X. If $\sum_{i=1}^{10}(x_i - p) = 3$ and $\sum_{i=1}^{10}(x_i - p)^2 = 9$ where $0 \ne p \in R$, then the standard deviation of these observation is : **[JEE(Main)Sept-2020]**

माना यादृच्छिक चर X के दस प्रेक्षण $x_i (1 \le i \le 10)$ है। यदि If $\sum_{i=1}^{10}(x_i - p) = 3$ तथा $\sum_{i=1}^{10}(x_i - p)^2 = 9$, जबकि $0 \ne p \in R$ है, तो इन प्रेक्षणों का मानक विचलन है : **[JEE(Main)Sept.2020]**

(A) $\sqrt{\frac{3}{5}}$ (A) $\frac{4}{5}$ (C) $\frac{9}{10}$ (D) $\frac{7}{10}$

Ans. C

Sol. $\sum_{i=1}^{10}(x_i - p) = 3$

$\sum_{i=1}^{10}(x_i - p)^2 = 9$

If we add or subtract a particular number in all deta then S.D remains unchanged.

hence $\sum_{i=1}^{10} x_i = 3$

$\sum_{i=1}^{10} x^2{}_i = 9$

$\sigma^2 = \frac{1}{N}\sum_{i=1}^{10} x_i^2 - \left(\bar{x}\right)^2$

$\sigma^2 = \frac{1}{10}(9) - \left(\frac{3}{10}\right)^2$

$\sigma^2 = \frac{90-9}{100}$

$S.D. = \sigma = \frac{9}{10}$

31. The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5, 7, 10, 12, 14, 15, then the absolute difference of the remaining two observations is : **[JEE(Main)Sept-2020]**

8 प्रेक्षणों का माध्य तथा प्रसारण क्रमशः 10 तथा 13.5 हैं। यदि इनमें से 6 प्रेक्षण 5, 7, 10, 12, 14 तथा 15 हैं, तो बाकी दो प्रेक्षणों का निरपेक्ष अंतर हैं : **[JEE(Main)Sept.2020]**

(A) 5 (B) 9 (C) 3 (D) 7

Ans D

Sol. Let remaining two observations are a and b.

$\text{Mean} = 10 = \frac{5+7+12+10+15+14+a+b}{8}$

$\Rightarrow 80 = 63 + a + b$

$a + b = 17$ .....(I)

Variance $= \sigma^2 = \frac{1}{n}\sum x_i^2 - \left(\bar{x}\right)^2$

$\Rightarrow 13.5 = \frac{1}{8}(25 + 49 + 144 + 100 + 225 + 196 + a^2 + b^2) - 100$

$\Rightarrow 108 = 739 + a^2 + b^2 - 800$

$\Rightarrow 908 = 739 + a^2 + b^2$

$\Rightarrow a^2 + b^2 = 169$

$\Rightarrow (a + b)^2 - 2ab = 169$

$\Rightarrow 289 - 169 = 2ab$ [use (I)]

$ab = 60$

$(a - b)^2 = (a + b)^2 - 4ab$

$\Rightarrow (a - b)^2 = 289 - 240$

$\Rightarrow (a - b)^2 = 49$

$|a - b| = 7$

32. If the variance of the following frequency distribution :

Class	:	10 – 20	20 – 30	30 –40
Frequencey	:	2	x	2

is 50, then x is equal to _____. **[JEE(Main)Sept-2020]**

यदि निम्न बारंबारता बंटन

वर्ग	:	10 – 20	20 – 30	30 –40
बारंबारता	:	2	x	2

का प्रसरण 50 है, तो x का मान है _____. **[JEE(Main)Sept.2020]**

Ans. 4

Sol.

x_i	5	15	25
f_i	2	x	2

$$\vec{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{10 + 15x + 50}{4 + x}$$

$$= \frac{15(4 + x)}{4 + x} = 15$$

$$\sigma^2 = 50 = \frac{\sum f_i x_i^2}{\sum f_i} - \left(\bar{x}\right)^2 = 50$$

$$\frac{50 + 225x + 1250}{4 + x} - (15)^2 = 50$$

$x = 4$

$$\therefore |a - b| = 7$$

33. The mean and variance of 7 observations are 8 and 16, respectively. If five observations are 2, 4, 10, 12, 14, then the absolute difference of the remaining two observations is : **[JEE(Main)Sept-2020]**

7 प्रेक्षणों का माध्य तथा प्रसारण क्रमशः 8 तथा 16 हैं। यदि पांच प्रेक्षण 2, 4, 10, 12, 14 हैं, तो शेष दो प्रेक्षणों का निरपेक्ष अंतर हैं : **[JEE(Main)Sept.2020]**

(A) 4 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Let the remaining observations are x and y.

$$\text{Mean} = \frac{2 + 4 + 10 + 12 + 14 + x + y}{7} = 8$$

$\Rightarrow 42 + x + y = 56 \Rightarrow x + y = 14$ ..... (1)

$$\text{Variance} = \frac{\sum_{i=1}^{7} x_i^2}{7} - \left(\bar{x}\right)^2 = 16$$

$$\Rightarrow \frac{2^2 + 4^2 + 10^2 + 12^2 + 14^2 + x^2 + y^2}{7} - (8)^2 = 16$$

$\Rightarrow 460 + x^2 + y^2 = 560 \Rightarrow x^2 + y^2 = 100$ ..... (2)

$(1)^2 - 2 \quad \Rightarrow 2xy = 196 - 100 = 96$

$\Rightarrow xy = 48$ ..... (3)

From (1) and (3)

$$x + \frac{48}{x} = 14 \Rightarrow x^2 - 14x + 48 = 0$$

$\Rightarrow x = 6$ and $y = 8$ or $x = 8$ and $y = 6$

$\Rightarrow |x - y| = 2$

34. If the mean and the standard deviation of the data 3, 5, 7, a, b are 5 and 2 respectively, then a and b are the roots of the equation : **[JEE(Main)Sept-2020]**

यदि आँकड़ों 3, 5, 7, a, b का माध्य तथा मानक विचलन क्रमशः 5 तथा 2 हैं, तो a तथा b जिस समीकरण के मूल हैं, वह है:

[JEE(Main)Sept.2020]

(A) $x^2 - 10x + 19 = 0$ (B) $2x^2 - 20x + 19 = 0$

(C) $x^2 - 20x + 18 = 0$ (D) $x^2 - 10x + 18 = 0$

Ans. A

Sol. $\text{Mean} = \frac{5+3+7+a+b}{5} = 5$

$a + b = 10$

$\text{S.D} = \left[\frac{5^2+3^2+7^2+a^2+b^2}{5} - 5^2\right]^{\frac{1}{2}} = 2$

$\frac{83+(a+b)^2-2ab}{5} - 25 = 4$

$83 + (10)^2 - 2ab - 125 = 20$

$ab = 19$

equation

$x^2 - (a+b)x + ab = 0$

$x^2 - (10)x + 19 = 0$

35. If $\sum_{i=1}^{n}(x_i - a) = n$ and $\sum_{i=1}^{n}(x_i - a)^2 = na$, $(n, a > 1)$ then the standard deviation of n observation $x_1, x_2, ..., x_n$ is :

[JEE(Main)Sept-2020]

यदि $\sum_{i=1}^{n}(x_i - a) = n$ तथा $\sum_{i=1}^{n}(x_i - a)^2 = na$, $(n, a > 1)$ है, तो n प्रेक्षणों $x_1, x_2, ..., x_n$ का मानक विचलन है :

[JEE(Main)Sept.2020]

(A) $\sqrt{n(a-1)}$ (B) $n\sqrt{a-1}$ (C) $a - 1$ (D) $\sqrt{a-1}$

Ans. D

Sol. $\sigma^2 = \frac{\sum(x_i - a)^2}{n} - \left(\frac{\sum(x_i - a)}{n}\right)^2$

$\sigma^2 = \frac{na}{n} - \left(\frac{n}{n}\right)^2 = a - 1$

$\sigma = \sqrt{a-1}$

36. Consider the data on x taking the values 0, 2, 4, 8, ..., 2^n with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \ldots {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to _____. **[JEE(Main)Sept-2020]**

आंकड़े जिनमें x के मानों 0, 2, 4, 8, ..., 2^n की बारंबारता क्रमशः ${}^nC_0, {}^nC_1, {}^nC_2, \ldots {}^nC_n$ है, पर विचार कीजिए। यदि इन आंकड़ो का माध्य $\frac{728}{2^n}$ है, तो n बराबर है _____ **[JEE(Main)Sept.2020]**

Ans. 6

Sol.

x	0	2	4	8	.....	2^n
f	${}^nC_0$	${}^nC_1$	${}^nC_2$	${}^nC_3$	.....	${}^nC_n$

$$\text{Mean} = \frac{\sum x_i f_i}{\sum f_i} = \frac{\sum_{r=1}^{n} 2^r \, {}^nC_r}{\sum_{r=0}^{n} {}^nC_r}$$

$$\text{Mean} = \frac{3^n - {}^nC_0}{2^n} = \frac{728}{2^n}$$

$3^n - 1 = 728$

$3^n = 729$

$\Rightarrow n = 6$

37. If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____. **[JEE(Main)Feb-2021]**

यदि दस धन पूर्णांकों 1, 1, 1, ..., 1, k का प्रसरण 10 से कम है, तो k का अधिकतम संभावित मान है _____ **[JEE(Main)Feb.2021]**

Ans. 11

Sol. $\sigma^2 < 10$

$$\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2 < 10$$

$$\frac{9 + k^2}{10} - \left(\frac{9+k}{10}\right)^2 < 10$$

$$\Rightarrow 90 + 10k^2 - 81 - k^2 - 18k < 1000$$

$$\Rightarrow 9k^2 - 18k - 991 < 0$$

$$k^2 - 2k < \frac{991}{9}$$

$$(k-1)^2 < \frac{1000}{9}$$

$$\Rightarrow \frac{-10\sqrt{10}}{3} < k-1 < \frac{10\sqrt{10}}{3}$$

$$k < \frac{10\sqrt{10}}{3} + 1$$

$$k \leq 11$$

k = 11 is largest integral value satisfying given inequality.

38. Let $X_1, X_2, \ldots\ldots X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i-\alpha)=36$ and $\sum_{i=1}^{18}(X_i-\beta)^2=90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is_____. **[JEE(Main)Feb.2021]**

माना $X_1, X_2, \ldots\ldots X_{18}$ अठारह प्रेक्षण हैं, जिनके लिए $\sum_{i=1}^{18}(X_i-\alpha)=36$ तथा $\sum_{i=1}^{18}(X_i-\beta)^2=90$ हैं, जहाँ α तथा β भिन्न वास्तविक संख्याएँ हैं। यदि इन प्रेक्षणों का मानक विचलन 1 है, तो $|\alpha - \beta|$ का मान बराबर है _____

[JEE(Main)Feb-2021]

Ans. 4

Sol. $$\sum_{i=1}^{18}(x_i-\alpha)=36$$

$$\therefore \sum_{i=1}^{18}x_i - 18\alpha = 36$$

$$\sum_{i=1}^{18}x_i = 36+18\alpha$$

$$\sum_{i=1}^{18}(x_i-\beta)^2 = 90$$

$$\sum_{i=1}^{18}x_i^2 - 2\beta\sum_{i=1}^{18}x_i + 18\beta^2 = 90$$

$$\sum_{i=1}^{18}x_i^2 = 90 + 2\beta(36+18\alpha) - 18\beta^2$$

Variance of these observations

$$1^2 = \frac{1}{18}\sum_{i=1}^{18} x_i^{\,2} - \left(\frac{\sum x_i}{18}\right)^2$$

$\Rightarrow 1 = \frac{1}{18}\ (90 + 2\beta\ (36 + 18\alpha) - 18\beta^2) - (2 + \alpha)^2$

$\Rightarrow 1 = 5 + 2\beta\ (2 + \alpha) - \beta^2 - (\alpha + 2)^2$

$\Rightarrow (\alpha + 2)^2 + \beta^2 - 2\beta(\alpha + 2) = 4$

$\Rightarrow (\alpha + 2 - \beta)^2 = 4$

$\Rightarrow (\alpha - \beta + 2)^2 - 4 = 0$

$\Rightarrow (\alpha - \beta + 4)\ (\alpha - \beta) = 0$

$\because\ \alpha \neq \beta$

$\therefore\ \alpha - \beta + 4 = 0$

$\alpha - \beta = -4$

$\therefore\ |\alpha - \beta| = 4$

39. Let in a series of 2n observations, half of them are equal to a and remaining half are equal to – a. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20 respectively. Then the value of $a^2 + b^2$ is equal to : **[JEE(Main)March-2021]**

माना 2n प्रेक्षणों की एक श्रृंखला में, आधे a के बराबर हैं तथा शेष आधे –a के बराबर हैं। प्रत्येक प्रेक्षणों में एक अचर b जोड़ने पर नये समूह का माध्य तथा मानक विचलन क्रमशः 5 तथा 20 हैं। तो $a^2 + b^2$ का मान बराबर है : **[JEE(Main)Mar.2021]**

(A) 425 (B) 925 (C) 650 (D) 250

Ans. A

Sol. Let observations are denoted by xi

For $1 \le i < 2n$

$$\overline{x} = \frac{\Sigma x_i}{2n} = \frac{(a + a + ... + a) - (a + a + ... + a)}{2n}$$

$$\overline{x} = 0$$

$$\text{and } (\sigma_x)^2 = \frac{\Sigma (x_i)^2}{2n} - (\overline{x})^2 = \frac{a^2 + a^2 + ... + a^2}{2n} - 0 = a^2$$

$\Rightarrow \sigma_x = a$

Now, adding a constant 'b' then,

$\overline{y} = \overline{x} + b = 5$]

$\Rightarrow b = 5$

and $\sigma_y = \sigma_x$ (No change in S.D.)

$\Rightarrow a = 20$

$\Rightarrow a^2 + b^2 = 425$

40. The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is : **[JEE(Main)March-2021]**

एक स्कूल में 25 अध्यापकों की औसत आयु 40 वर्ष है। एक अध्यापक 60 वर्ष की आयु में सेवानिवृत्त होता है तथा उसकी जगह एक नये अध्यापक की नियुक्ति होती है। यदि अब इस स्कूल में अध्यापकों की औसत आयु 39 वर्ष है, तो नये नियुक्त किए गए अध्यापक की आयु (वर्षों में) है _____ **[JEE(Main)Mar.2021]**

Ans. 35

Sol. Sum of ages = $25 \times 40 = 1000$

Let the age of new teacher = x years

$$\frac{1000 - 60 + x}{25} = 39$$

$\Rightarrow x = 35$

41. Consider a set of 3n numbers having variance 4. In this set, the mean of first 2n numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first 2n numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then 9k is equal to _____.

[JEE(Main)March-2021]

3n संख्याओं का एक समुच्चय है, जिसका प्रसरण 4 है। इस समुच्चय में, प्रथम 2n संख्याओं का माध्य 6 है तथा शेष n संख्याओं का माध्य 3 है। प्रथम 2n संख्याओं में प्रत्येक में 1 जोड़कर तथा शेष n संख्याओं में प्रत्येक से 1 घटा कर एक नया समुच्चय बनाया गया है। यदि नये समुच्चय का प्रसरण k है, तो 9k बराबर है _____ **[JEE(Main)Mar.2021]**

Ans. 68

Sol. First αn number $\rightarrow a_1, a_2, a_3 \ldots\ldots a_{2n}$

Last n number $\rightarrow b_1, b_2, \ldots\ldots b_n$

$$\sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} - (\bar{x}) \qquad \bar{x} = \frac{2n \times 6 + n \times 3}{3n}$$

$$\sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} - 5^2 \qquad \bar{x} = 5$$

$$\mu = \frac{\sum a^2 + \sum b^2}{3n} - 25$$

$$\sum a^2 + \sum b^2 = 87\,n$$

Now first 2n numbers $\rightarrow a_1 + 1, a_2 + 1, \ldots\ldots\ldots a_{2n} + 1$

Last n numbers $\rightarrow b_1 - 1, b_2 - 1, \ldots\ldots\ldots b_n - 1$

$$\sigma^2 = \frac{\sum (a+1)^2 + \sum (b-1)^2}{3n} - (\bar{x})^2$$

$$\sigma^2 = \frac{\left(\sum a^2 + n + 2\sum a\right) + \left(\sum b^2 + n - 2\sum b\right)}{3n} - \left(\frac{16}{3}\right)^2$$

$$\sigma^2 = \frac{87n + 3n + 2(12n) - 2(3n)}{3n} - \left(\frac{16}{3}\right)^2$$

$$\sigma^2 = \left(\frac{108}{3}\right) - \left(\frac{16}{3}\right)^3$$

$$\sigma^2 = \frac{68}{9}$$

then Ans. $9k = 9\left(\frac{68}{9}\right) = 68$

42. Consider the statistics of two sets of observations as follows:

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to _____.

[JEE(Main)March-2021]

नीचे दी गई प्रेक्षणों के दो समूहों की सांख्यिकी का विचार कीजिए :

	आकार	माध्य	प्रसरण
प्रेक्षण I	10	2	2
प्रेक्षण II	n	3	1

यदि इन दोनों प्रेक्षणों को मिलाकर बने समूह का प्रसरण $\frac{17}{9}$ है, तो n का मान बराबर है _____। **[JEE(Main)Mar.2021]**

Ans. 5

Sol. For observation I :

$$\sum x_i = 10 \times 2 = 20$$

$$\sigma^2 = \frac{\Sigma x_i^2}{10} - (2)^2 = 2$$

$$\Sigma x_i^2 = 60$$

For observation II :

$$\Sigma x_i = n \times 3 = 3n$$

$$\sigma^2 = \frac{\Sigma x_i^2}{n} - (3)^2 = 1$$

$$\Sigma x_i^2 = 10n$$

For combined (n + 10) observations

$$\Sigma x_i = 20 + 3n$$

$$\Sigma x_i^2 = 60 + 10n$$

$$\sigma^2 = \left(\frac{60+10n}{n+10}\right) - \left(\frac{20+3n}{n+10}\right)^2 = \frac{17}{9}$$

n = 5

43. Consider three observations a, b and c such that b = a + c. If standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true ? **[JEE(Main)March-2021]**

तीन प्रेक्षणों a, b तथा c का विचार, जिनके, लिए b = a + c है। यदि a + 2, b + 2, c + 2 का मानक विचलन d है, तो निम्न में से कौनसा सत्य है ? **[JEE(Main)Mar.2021]**

(A) $b^2 = a^2 + c^2 + 3d^2$ (B) $b^2 = 3(a^2 + c^2) - 9d^2$

(C) $b^2 = 3(a^2 + c^2) + 9d^2$ (D) $b^2 = 3(a^2 + c^2 + d^2)$

Ans. B

Sol. $AM = \bar{x} = \frac{a+b+c}{3} = \frac{2b}{3}$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$d^2 = \left(\frac{a^2+b^2+c^2}{3}\right) - \left(\frac{2b}{3}\right)^2$$

$9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$

$b^2 = 3a^2 + 3c^2 - 9d^2$

44. The mean of 6 distinct observations is 6.5 and their variance is 10.25. If 4 out of 6 observations are 2, 4, 5 and 7, then the remaining two observations are : **[JEE(Main)July2021]**

6 भिन्न प्रेक्षणों का माध्य 6.5 है तथा उनका प्रसरण 10.25 है। यदि 6 प्रेक्षणों में से 4 प्रेक्षण 2, 4, 5 तथा 7 है, तो शेष दो प्रेक्षण है : **[JEE(Main)July2021]**

(A) 3, 18 (B) 1, 20 (C) 10, 11 (D) 8, 13

Ans. C

Sol. $\bar{x} = 6.5$

$$\frac{18+x+y}{6} = 6.5$$

$\Rightarrow x + y = 21$ .....(i)

$\sigma^2 = 10.25$

$$\frac{\Sigma x^2}{n} - (\bar{x})^2 = 10.25$$

$$\frac{94+x^2+y^2}{6} - (6.5)^2 = 10.25$$

$x^2 + y^2 = 221$

$(x + y)^2 - 2xy = 221$

$xy = 110$ .....(ii)

solving equation (i) & (ii)

$x = 10, y = 11$

45. If the mean and variance of six observations 7, 10, 11, 15, a, b are 10 and $\frac{20}{3}$, respectively, then the value of $|a-b|$ is equal to : **[JEE(Main)July2021]**

यदि 6 प्रक्षणों 7, 10, 11, 15, a तथा b का माध्य तथा प्रसरण क्रमशः 10 तथा $\frac{20}{3}$ हैं, तो $|a-b|$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 11 (B) 1 (C) 7 (D) 9

Ans B

Sol. $\text{mean} = \frac{7+10+11+15+a+b}{6} = 10$

$4 + b = 17$

$\text{variance} = \frac{7^2+10^2+11^2+15^2+a^2+b^2}{6} - 100 = \frac{20}{3}$

$a^2 + b^2 = 145$

$(a + b)^2 - 2ab = 145$

$ab = 72$

$(a-b)^2 = (a+b)^2 - 4ab$

$|a-b| = 1$

46. The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviaton $\sqrt{13.44}$, then the standard deviation of the second sample is: **[JEE(Main)July2021]**

एक समूह के दो नमूनों में से पहले नमूनें में 100 वस्तुएँ हैं जिनका माध्य 15 तथा मानक विचलन 3 हैं। यदि पूरे समूह में 250 वस्तुएँ हैं और उनका माध्य 15.6 तथा मानक विचलन $\sqrt{13.44}$ हैं, तो दूसरे नमूने का मानक विचलन हैं : **[JEE(Main)July2021]**

(A) 6 (B) 4 (C) 5 (D) 8

Ans B

Sol. $15.6 = \frac{100\times15+150\times\bar{x}}{250}$

$\bar{x} = 16$

Combined satndard deviation $= \sqrt{13.44}$

$\Rightarrow$ Combined variance $(\sigma^2) = 13.44$

$$\sigma^2 = \frac{\Sigma x_i^2}{n} - (\bar{x})^2$$

$$13.44 = \frac{\Sigma x_i^2}{250} - 243.36$$

$$\Rightarrow \Sigma x_i^2 = 64200$$

For 1st sample

$$9 = \frac{\Sigma x_I^2}{100} - 225$$

$$\Rightarrow \Sigma x_I^2 = 23400$$

For 2nd sample

$$\Sigma x_{II}^2 = 64200 - 23400 = 40800$$

standard deviation of IInd sample will be

$$\sqrt{\frac{\Sigma x_{II}^2}{n} - (\bar{x}_{II})^2} = \sqrt{\frac{40800}{150} - 256} = 4$$

47. If the mean and variance of the following data:

6, 10, 7, 13, a, 12, b, 12

are 9 and $\frac{37}{4}$ respectively, then $(a-b)^2$ is equal to : **[JEE(Main)July2021]**

यदि आँकड़ों : 6, 10, 7, 13, a, 12, b, 12 का माध्य तथा प्रसरण क्रमशः 9 तथा $\frac{37}{4}$ हैं, तो $(a-b)^2$ बराबर है :

[JEE(Main)July2021]

(A) 12 (B) 24 (C) 16 (D) 32

Ans C

Sol. $$\bar{x} = \frac{6+10+7+13+a+12+b+12}{8} = 9$$

$$a + b = 12$$

$$\sigma^2 = \frac{\sum xi^2}{8} - (\bar{x})^2 = \frac{37}{4}$$

$$\Rightarrow \frac{6^2+10^2+7^2+13^2+a^2+12^2+b^2+12^2}{8} - 81 = \frac{37}{4}$$

$\Rightarrow a^2 + b^2 = 80$

$(a - b)^2 + (a + b)^2 = 2(a^2 + b^2)$

$(a - b)^2 = 160 - 144 = 16$

48. Let the mean and variance of the frequency distribution

x:	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
f:	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be :

[JEE(Main)July2021]

माना बारंबारता बंटन

x:	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
f:	4	4	α	β

के माध्य तथा प्रसरण क्रमशः 6 तथा 6.8 हैं। यदि x_3 को 8 से 7 कर दिया जाए, तो नये आँकड़ों का माध्य होगा :

[JEE(Main)July2021]

(A) 5 (B) 4 (C) $\frac{16}{3}$ (D) $\frac{17}{3}$

Ans D

Sol. $\mu = \frac{\sum f_i x_i}{\sum f_i}$

$\Rightarrow \mu = \frac{2\times4+6\times4+8\times\alpha+9\times\beta}{4+4+\alpha+\beta}$

$\Rightarrow 6 = \frac{32+8\alpha+9\beta}{8+\alpha+\beta}$

$\Rightarrow 48 + 6\alpha + 6\beta = 32 + 8\alpha + 9\beta$

$\Rightarrow 2\alpha + 8\beta = 16$ .....(i)

$\sigma^2 = \frac{\sum f_i.(x_i - \mu)^2}{\sum f_i}$

$\Rightarrow 6.8 = \frac{4\times16+4\times0+\alpha\times4+\beta\times9}{8+\alpha+\beta}$

$\Rightarrow \frac{34}{5} = \frac{4+4\alpha+9\beta}{8+\alpha+\beta}$

$\Rightarrow 14\alpha - 11\beta = 48$ .....(ii)

On solving (i) & (ii)

$\alpha = 5$ and $\beta = 2$

New mean (μ) =

$$\frac{4\times 2+4\times 6+\alpha\times 7+\beta\times 9}{8+\alpha+\beta}$$

$$=\frac{8+24+5\times 7+2\times 9}{8+5+2}$$

$$=\frac{85}{15}=\frac{17}{3}$$

49. Consider the following frequency distribution :

Class :	0 –6	6 –12	12– 18	18 – 24	24 – 30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then the value $(a-b)^2$ is equal to. **[JEE(Main)July2021]**

निम्न बारंबारता बंटन पर विचार कीजिए :

वर्ग :	0 – 6	6 –12	12– 18	18 – 24	24 – 30
बारंबारता :	a	b	12	9	5

यदि इसका माध्य $\frac{309}{22}$ तथा माध्यिका 14 हैं, तो $(a-b)^2$ बराबर है _____ **[JEE(Main)July2021]**

Ans D

Sol.

Class	x_i	f_i	f_ix_i
0 – 6	3	a	3a
6 – 12	9	b	9b
12 – 18	15	12	180
18 – 24	21	9	189
24 – 30	27	5	135

$$\text{Mean} = \frac{3a+9b+180+189+135}{a+b+12+9+5}=\frac{309}{22}$$

$18a + 37b = 1018$ .....(1)

$$\text{Now median} = 12+\left(\frac{\frac{a+b+26}{2}-\quad a+b}{12}\right)\times 6=14$$

$a + b = 18$ .....(2)

by (1) & (2)

$a = 8, \quad b = 10$

$(a - b)^2 = 4$

50. Consider the following frequency distribution :

Class:	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency:	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to _____.

[JEE(Main)July2021]

निम्न बारंबारता बंटन पर विचार कीजिए :

वर्ग :	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
बारंबारता :	α	110	54	30	β

यदि सभी बारंबारताओं का योग 584 है। तथा माध्यिका 45 है, तो $|\alpha - \beta|$ बराबर है _____

[JEE(Main)July2021]

Ans. 164

Sol.

class	frequency	cumulative freq.
10 – 20	α	α
20 – 30	110	$\alpha + 110$
30 – 40	54	$\alpha + 164$
40 – 50	30	$\alpha + 194$
50 – 60	β	$\alpha + \beta + 194 = 584$

$N = 584$

$\alpha + \beta = 390$

$\frac{N}{2} = 292$

$$\text{Median} = 45 = \ell + \left[\frac{\frac{N}{2} - F}{f}\right] \times 4$$

$$45 = 40 + \left[\frac{292 - (\alpha + 164)}{30}\right]$$

$\alpha = 113$

$\beta = 277$

$|\alpha - \beta| = 164$

51. The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value owas taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is : **[JEE(Main)Aug.2021]**

20 प्रेक्षणों के माध्य तथा मानक विचलन क्रमशः 10 तथा 2.5 निकाले गये। यह पाया गया कि गलती से एक आंकड़ा 35 की जगह 25 लिया गया था। यदि सही आंकड़ों का माध्य तथा मानक विचलन क्रमशः α तथा $\sqrt{\beta}$ हैं, तो (α, β) है : **[JEE(Main)Aug.2021]**

(A) (11, 25) (B) (10.5, 25) (C) (11, 26) (D) (10.5, 26)

Ans D

Sol. $n = 20 \qquad \bar{x} = 10 \qquad \sigma = 2.5$

$\Sigma\bar{x} = n\bar{x} = 200$

$$\sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

$$6.25 = \frac{\Sigma x_i^2}{10} - 100$$

$$\frac{\Sigma x_i^2}{20} = 106.25$$

$\Sigma x_i^2 = 2125$

For new distribution

$$\Sigma x_i = 200 - 25 + 35 = 210 \quad \Rightarrow \quad \bar{x} = \frac{\Sigma x_i}{20} = \frac{210}{20} = 10.5$$

$\alpha = 10.5$

$\Sigma x_i^2 = 2125 - (25)^2 + (35)^2 = 2725$

$$\beta = \sigma^2 = \frac{\Sigma x_i^2}{20} - (\bar{x})^2 = \frac{2725}{20} - (10.5)^2$$

$$= 136.25 - 110.25$$

$$= 26$$

$(\alpha, \beta) \equiv (10.5, 26)$

52. The mean and variance of 7 observations are 8 and 16 respectively. If two observations are 6 and 8, then the variance of the remaining 5 observations is : **[JEE(Main)Aug.2021]**

7 प्रेक्षणों का माध्य तथा प्रसरण क्रमशः 8 तथा 16 हैं। यदि दो प्रेक्षण 6 तथा 8 हैं, तो शेष 5 प्रेक्षणों का प्रसरण है : **[JEE(Main)Aug.2021]**

(A) $\frac{134}{5}$ (B) $\frac{112}{5}$ (C) $\frac{536}{25}$ (D) $\frac{92}{5}$

Ans C

Sol. $x_1, x_2, x_3, x_4, x_5, 8, 6$

$$\frac{x_1+x_2+x_3+x_4+x_5+14}{7}=8$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 42 \ldots (1)$$

$$\frac{x_1^2+x_2^2+x_3^2+x_4^2+x_5^2+8^2+6^2}{7}-64=16$$

$$x_1^2+x_2^2+x_3^2+x_4^2+x_5^2=460$$

So variance of $x_1, x_2 \ldots\ldots x_5$

$$=\frac{460}{5}-\left(\frac{42}{5}\right)^2=\frac{536}{25}$$

53. Let the mean and variance of the four numbers 3, 7, x and y (x > y) be 5 and 10 respectively. Then the mean of four numbers 3 + 2x, 7 + 2y, x + y and x – y is _____. **[JEE(Main)Aug.2021]**

माना चार संख्याओं 3, 7, x तथा y (x > y) के माध्य तथा प्रसरण क्रमशः 5 तथा 10 है तो चार संख्याओं 3 + 2x, 7 + 2y, x + y तथा x – y का माध्य है_____ **[JEE(Main)Aug.2021]**

Ans 12

Sol. $\frac{3+7+x+y}{4}=5$

$$x + y = 10 \qquad \text{_____(1)}$$

$$\sigma^2=10=\frac{3^2+7^2+x^2+y^2}{4}-(5)^2$$

$$x^2 + y^2 = 82 \qquad \text{_____(2)}$$

Using equ. (1) and (2)

$x = 9, y = 1$

Four numbers are 21, 9, 10, 8

$$\text{mean} = \frac{21+9+10+8}{4} = 12$$

54. Let n be an odd natural number such that the variance of 1, 2, 3, 4...., n is 14. Then n is equal to_____.

[JEE(Main)Aug.2021]

माना n एक विषम प्राकृतिक संख्या है जिसके लिए 1, 2, 3, 4...., n का प्रसरण 14 है तो n बराबर है_____

[JEE(Main)Aug.2021]

Ans 13

Sol. $\sigma^2 = \left(\frac{\Sigma n^2}{n}\right) - \left(\frac{\Sigma n}{n}\right)^2 = 14$

$$\Rightarrow \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = 14$$

$$\Rightarrow n^2 = 169 \Rightarrow n = 13$$

55. An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu+\sigma^2$ is equal to _____. **[JEE(Main)Aug.2021]**

50 परीक्षार्थियों द्वारा एक ऑनलाइन परीक्षा दी गई। इन परीक्षार्थियों में 20 लड़के द्वारा प्राप्त किए गए अंको का माध्य 12 तथा प्रसरण 2 हैं। 30 लड़कियों द्वारा प्राप्त अंको का प्रसरण भी 2 है। सभी 50 परीक्षार्थियों के अंको का माध्य 15 है। यदि लड़कियों के अंको का माध्य μ है तथा 50 परीक्षार्थियों के अंको का प्रसरण σ^2 है, तो $\mu+\sigma^2$ बराबर है_____ **[JEE(Main)Aug.2021]**

Ans 25

Sol. $\sigma_b^2 = 2$ $\quad n_1 = \text{No of boys} = 20$

$\bar{x}_b = 12$ $\quad n_2 = \text{No. of girls} = 30$

$\sigma_g^2 = 2$

$\bar{x}_g = \mu$

$\bar{x} = 15$

Now $\bar{x} = \frac{\overline{x_b} n_1 + \bar{x}_g n_2}{n_1 + n_2}$

$15 = \frac{12 \times 20 + \bar{x}_g \times 30}{30 + 20}$

$750 - 240 = 30\ \bar{x}_g$

$\bar{x}_g = 17$

$\mu = 17$

variance of combined group

$\sigma^2 = \frac{n_1\sigma_b^2 + n_2\sigma_g^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_b - \bar{x}_g)^2$

$\sigma^2 = \frac{20 \times 2 + 30 \times 2}{20 + 30} + \frac{20 \times 30}{(20 + 30)^2}(12 - 17)^2$

$\sigma^2 = 8$

$\Rightarrow \mu + \sigma^2 = 17 + 8$

$= 25$

56. The mean of 10 numbers $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, ...$is _____. **[JEE(Main)Aug.2021]**

10 संख्याओं $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, ...$का माध्य है _____ **[JEE(Main)Aug.2021]**

Ans 398

Sol. $T_r = (3r + 4).(2r + 6)$

$T_r = 6r^2 + 26r + 24$

$\bar{x} = \frac{\sum_{r=1}^{10} T_r}{10} = \frac{6.\frac{10 \times 11 \times 21}{6} + 26.\frac{10 \times 11}{2} + 24 \times 10}{10}$

$\bar{x} = 398$

57. Let X be a random variable with distribution.

x	-2	-1	3	4	6
$P(X = x)$	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2, then $100\ \sigma^2$ is equal to : **[JEE(Main)Aug.2021]**

माना एक यादृच्छया चर X का बंटन निम्न है :

x	-2	-1	3	4	6
$P(X=x)$	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

यदि X का माध्य 2.3 है तथा X का प्रसरण σ^2 है, तब 100 σ^2 बराबर है _____ **[JEE(Main)Aug.2021]**

Ans 781

Sol. $\Sigma P_i = 1$

$$\Rightarrow \frac{1}{5}+a+\frac{1}{3}+\frac{1}{5}+b=1$$

$$\Rightarrow a+b=1-\left(\frac{2}{5}+\frac{1}{3}\right)$$

$$\Rightarrow a+b=1-\frac{11}{15}$$

$$\therefore a+b=\frac{4}{15} \quad .....(1)$$

$$\mu=\Sigma p_i x_i$$

$$\Rightarrow 2.3=\frac{-2}{5}-a+1+\frac{4}{5}+6b$$

$$a-6b=1+\frac{2}{5}-\frac{23}{10}$$

$$a-6b=\frac{10+4-23}{10}$$

$$a-6b=\frac{-9}{10} \quad .....(2)$$

Solving the equatino (1) and (2)

$$a=\frac{1}{10},\ b=\frac{1}{6}$$

using the formula $\sigma^2=\Sigma p_i x_i^2-(\bar{x})^2$

$$=\frac{1}{5}\times 4+\frac{1}{10}.(1)+\frac{1}{3}\times 9+\frac{1}{5}\times 16+\frac{1}{6}\times 36$$

$-(2.3)^2$

$$=\frac{4}{5}+\frac{1}{10}+9+\frac{16}{5}-\left(\frac{23}{10}\right)^2$$

$100\sigma^2 = 781$

58. The mean of the number a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to : **[JEE(Main)June2022]**

संख्याओं a, b, 8, 5, 10 का माध्य 6 है तथा इनका प्रसरण 6.8 है। यदि इन संख्याओं का माध्य के सापेक्ष माध्य विचलन M है, तो 25 M बराबर है : **[JEE(Main)June2022]**

(A) 60 (B) 55 (C) 50 (D) 45

Ans. A

Sol. $$\sigma^2 = \frac{\sum_{i=1}^{5}(x_i-\bar{x})^2}{5}$$

Mean = 6

$$\frac{a+b+8+5+10}{5}=6$$

$a + b = 7$

$b = 7 - a$

$$6.8=\frac{(a-6)^2+(b-6)^2+(8-6)^2+(5-6)^2+(10-6)^2}{5}$$

$34 = (a-6)^2 + (7-a-6)^2 + 4 + 1 + 16$

$a^2 - 7a + 12 = 0 \Rightarrow a = 4$ or $a = 3$

$a = 4$	$a = 3$
$b = 3$	$b = 4$

$$M=\frac{\sum_{i=1}^{5}|x_i-x|}{5}$$

$$M=\frac{|a-6|+|b-6|+|8-6|+|5-6|+|10-6|}{5}$$

when $a = 3, b = 4$	when $a = 4, b = 3$
$M=\frac{3+2+2+1+4}{5}$	$M=\frac{2+3+2+1+7}{5}$
$M=\frac{12}{5}$	$M=\frac{12}{5}$

$$25M = 25 \times \frac{12}{5} = 60$$

59. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to : **[JEE(Main)June2022]**

50 प्रेक्षणों के माध्य तथा मानक विचलन क्रमशः 15 तथा 2 हैं। यह पाया गया कि एक गलत प्रेक्षण लिया गया था तथा सही और गलत प्रेक्षणों का योग 70 है। यदि सही माध्य 16 है, तो सही प्रसरण बराबर है : **[JEE(Main)June2022]**

(A) 10 (B) 36 (C) 43 (D) 60

Ans. C

Sol. Given $\overline{x} = 15$

$\sigma = 2 \Rightarrow \sigma^2 = 4$

$x_1 + x_2 + ... + x_{50} = 15 \times 50 = 750$

$$4 = \frac{x_1^2 + x_2^2 + ... + x_{50}^2}{50} - 225$$

$x_1^2 + x_2^2 + ... + x_{50}^2 = 50 \times 229$

Let a be the correct observation and b is the incorrect observation

a + b = 70

$$16 = \frac{750 - b + a}{50}$$

a – b = 50

= a = 60 b = 10

correct variance $\frac{50 \times 229 + (60)^2 - (10)^2}{50} - 256$

= 43

Let a be the correct observation and b is the incorrect observation

a + b = 70

60. The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y, where x < y, are 6 and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to : **[JEE(Main)June2022]**

आंकड़ों 4, 5, 6, 6, 7, 8, x, y, जहाँ x < y है, के माध्य तथा प्रसरण क्रमशः 6 तथा $\frac{9}{4}$ हैं। तो $x^4 + y^2$ बराबर है : **[JEE(Main)June2022]**

(A) 162 (B) 320 (C) 674 (D) 420

Ans. B

Sol. $\frac{36+x+y}{8}=6 \quad \Rightarrow \frac{9}{4}=\frac{16+25+36+36+49+64+x^2+y^2}{8}-(6)^2$

$x+y=12 \quad \Rightarrow x^2+y^2=306-226$

$\Rightarrow x^2+y^2=80$

$x=4, y=8$

$x^4+y^2=256+64 \quad \Rightarrow 320$

61. Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observation are $\frac{7}{2}$ and a respectively, the $(4a + x_5)$ is equal to :

[JEE(Main)June2022]

माना 5 प्रेक्षणों x_1, x_2, x_3, x_4, x_5 के माध्य तथा प्रसरण क्रमशः $\frac{24}{5}$ तथा $\frac{194}{25}$ हैं। यदि प्रथम 4 प्रेक्षणों के माध्य तथा प्रसरण क्रमशः $\frac{7}{2}$ तथा a हैं, तो $(4a + x_5)$ बराबर है : **[JEE(Main)June2022]**

(A) 13 (B) 15 (C) 17 (D) 18

Ans. B

Sol. $\text{Mean}=\frac{x_1+x_2+x_3+x_4+x_5}{5}=\frac{24}{5} \Rightarrow x_1+x_2+x_3+x_4+x_5=24$

$\Rightarrow \frac{x_1+x_2+x_3+x_4}{4}=\frac{7}{2}$

$\Rightarrow x_1+x_2+x_3+x_4=14 \qquad \Rightarrow x_5=10$

$\frac{\sum_{i=1}^{5} x_1^2}{5}-\left(\frac{24}{5}\right)^2=\frac{194}{25}$

$\sum_{i=1}^{5} x_i^2=154 \Rightarrow x_1^2+x_2^2+x_3^2+x_4^2=154-100=54$

$\frac{\sum_{i=1}^{4} x_1^2}{4}-\left(\frac{7}{2}\right)^2=a$

$\Rightarrow a=\frac{5}{4}$

$x_5 + 4a = 10 + 5 = 15$

62. The number of values of $a \in N$ such that the variance of 3, 7, 12, a, 43 – a is a natural number is :

[JEE(Main)June2022]

$a \in N$ के मानों की संख्या 3, 7, 12, a, 43 – a का प्रसरण एक धन पूर्णांक है : **[JEE(Main)June2022]**

(A) 0 (B) 2 (C) 5 (D) infinite (अनन्त)

Ans. A

Sol. Given numbers : 3, 7, 12, a, 43 – a

$$\text{Mean} = \overline{x} = \frac{3+7+12+a+43-9}{5} = 13$$

$$\text{Variance} = \frac{9+49+144+a^2+(43-a)^2}{5} - 13^2$$

So, variance $\in$ natural number

$$\frac{2a^2-a+1}{5} \in N$$

$$2a^2-a+1 = 5N$$

$$2a^2-a+1-5N = 0$$

It is quadratic in a

$$a = \frac{1 \pm \sqrt{1-4(2)(1-5N)}}{4} = \frac{1 \pm \sqrt{40N-7}}{4}$$

So, Δ cannot be perfect square & in the form of 4λ or $4\lambda+$.

So, a cannot be perfect square

$\therefore$ Number of values = 0

63. If the mean deviation about the mean of the number 1, 2, 3, ..., n where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.

[JEE(Main)June2022]

यदि संख्याओं 1, 2, 3, ...n, जहाँ n विषम है, का माध्य के सापेक्ष माध्य विचलन $\frac{5(n+1)}{n}$ है, तो n बराबर है _____

[JEE(Main)June2022]

Ans. 21

Sol. $$\overline{x} = \frac{1+2+3+.....+n}{n} = \frac{n+1}{2}$$

$$\sum |x_i - \overline{x}| = |1-\overline{x}| + |2-\overline{x}| + ... + \left|\frac{n+1}{2} - \overline{x}\right| + ... + |n-\overline{x}|$$

$$= 2.\left\{\left|1-\overline{x}\right|+\left|2-\overline{x}\right|+.....+\left|\frac{n-1}{2}-\overline{x}\right|\right\}$$

$$= 2\left\{\left|\overline{x}-1\right|+\left|\overline{x}-2\right|+.....+\left(\overline{x}-\frac{n-1}{2}\right)\right\}$$

$$= 2\left\{\frac{n-1}{2}.\overline{x}-\frac{\left(\frac{n-1}{2}.\frac{n+1}{2}\right)}{2}\right\}$$

$$= \frac{(n-1).(n+1)}{2}-\frac{(n-1).(n+1)}{4}$$

$$\sum\left|x_i-\overline{x}\right| = \frac{n^2-1}{4}$$

$$\frac{\sum\left|x_i-\overline{x}\right|}{n} = \frac{n^2-1}{4n} = \frac{5n+5}{n}$$

$\Rightarrow$ $n^2 - 20n - 21 = 0$

$(n + 1)(n - 21) = 0$

$n = 21$

64. The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

[JEE(Main)June2022]

15 प्रेक्षणों के माध्य तथा मानक विचलन क्रमशः 8 तथा 3 ज्ञात किए गए। पुनः जाँच करने पर पाया गया कि प्रेक्षण 20 को गलती से 5 पढ़ा गया था। तो सही प्रसरण बराबर है _____. **[JEE(Main)June2022]**

Ans. 17

Sol. We have

Variance $\frac{\sum_{r=1}^{15} X_r^2}{15}-\left(\frac{\sum_{r=1}^{15} X_r}{15}\right)^2$

Now, as per information given in equation $\frac{\sum X_r^2}{15}-8^2 = 3^2 \Rightarrow \sum X_r^2 = 1095$

Now, the new $\sum X_r = (15\times 8)-5+(20) = 135$

$$\therefore \text{ Variance } = \frac{1470}{15} - \left(\frac{135}{15}\right)^2 = 98 - 81 = 17$$

65. Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62 and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is _____. **[JEE(Main)June2022]**

माना एक कक्षा में 7 छात्र हैं। गणित की परीक्षा में इन छात्रों के अंकों का माध्य 62 तथा प्रसरण 20 हैं। कोई छात्र परीक्षा में फेल होता है। यदि वह 50 अंकों से कम प्राप्त करता है, तो फेल होने वाले छात्रों की अधिकतम संख्या है_____. **[JEE(Main)June2022]**

Ans. 0

Sol. According to given data

$$\bar{x} = 62, \frac{\sum(x_i - \bar{x})^2}{7} = 20$$

$\Rightarrow (x_1 - 62)^2 + (x_2 - 62)^2 + ...(x_7 - 62)^2 = 140$ for failure of any student $x_i < 50$

Suppose $x_1 = 49$.

$(49 - 62)^2 = 169$ which is not possible. Therefore no student is going to score less. than 50.

= 0 students

66. If the mean deviation about median for the numbers 3, 5, 7, 2k, 12, 16, 21, 24, arranged in the ascending order, is 6 then the median is : **[JEE(Main)July2022]**

यदि आरोही क्रम में लिखी संख्याओं 3, 5, 7, 2k, 12, 16, 21, 24 का माध्यिका के सापेक्ष माध्य विचलन 6 है, तो माध्यिका है : **[JEE(Main)July2022]**

(A) 11.5 (B) 10.5 (C) 12 (D) 11

Ans. D

Sol. $\text{Median} = \frac{2k+12}{2} = k+6$

$$\text{M.D.} = \frac{\Sigma|x_i - M|}{n} = 6$$

$$= \frac{(k+3)+(k+1)+(k-1)+(6-k)+(6-k)+(10-k)+(15-k)+(18-k)}{8}$$

$$\Rightarrow \frac{58-2k}{8} = 6$$

$$\Rightarrow k = 5$$

Median = 5 + 6 = 11

67. The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $P(X = 1) = \frac{4}{243}$, then $P(X = 4 \text{ or } 5)$ is equal to : **[JEE(Main)July2022]**

एक द्विपद बंटन के माध्य तथा प्रसरण क्रमशः α तथा $\frac{\alpha}{3}$ हैं। यदि $P(X = 1) = \frac{4}{243}$ है, तो $P(X = 4$ या $5)$ बराबर है :

[JEE(Main)July2022]

(A) $\frac{5}{9}$ (B) $\frac{64}{81}$ (C) $\frac{16}{27}$ (D) $\frac{145}{243}$

Ans. C

Sol. $np = \alpha$ .....(1)

$npq = \alpha/3$ .....(2)

From (1) & (2)

$q = 1/3$ & $p = 2/3$

$^{n}C_1 q^{n-1} p^1 = \frac{4}{243}$

$\frac{n}{3^n} = \frac{2}{243}$

$n = 6$

$P(4 \text{ or } 5) = {}^{6}c_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^{6}C_5 \left(\frac{2}{3}\right)^5 \cdot \left(\frac{1}{3}\right)^0$

$= \frac{16}{27}$

68. The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____. **[JEE(Main)July2022]**

40 प्रेक्षणों का माध्य तथा मानक विचलन क्रमशः 30 तथा 5 हैं। यह पाया गया कि इनमें से दो प्रेक्षण 12 तथा 10 गलती से लिखे गए। यदि गलती से लिखे दो प्रक्षेणों को हटाने के पश्चात् शेष आंकड़ों का मान विचलन σ है, तो $38\sigma^2$ बराबर है_____।

[JEE(Main)July2022]

Ans. 238

Sol. Wrong mean $= \mu_1 = 30$

Wrong S.D $= \sigma_1 = 5$

$\frac{\Sigma x_i}{40} = 30$

$\Rightarrow \Sigma x_i = 1200$

$\sigma_1^2 = 25$

$\Rightarrow \frac{\Sigma x_i^2}{40} - 30^2 = 25$

$\Rightarrow \Sigma x_i^2 = 925 \times 40 = 37000$

New sum $= \Sigma x_i' = 1200 - 10 - 12 = 1178$

New mean $= \mu_i' = \frac{1178}{38} = 31$

New $\Sigma x_i^2 = 37000 - (10)^2 - (12)^2 = 36756$

New S.D, $\sigma_1' = \sqrt{\frac{36756}{38} - (31)^2} = \sigma$

$36756 - (31)^2 \times 38 = 38\sigma^2$

$\Rightarrow 38\sigma^2 = 238$

69. The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is _____.

[JEE(Main)July2022]

एक विद्यार्थी द्वारा 10 प्रेक्षणों के माध्य तथा प्रसरण क्रमशः 15 तथा 15 निकाले गए। विद्यार्थी ने एक परीक्षण 15 को गलती से 25 लिया। तो सही मानक विचलन है _____. **[JEE(Main)July2022]**

Ans. 2

Sol. $n = 10, \bar{x} = \frac{\sum x_i}{10} = 15$

$6^2 = \frac{\sum x_i^2}{10} - (\bar{x})^2 = 15$

$\Rightarrow \sum_{i=1}^{10} x_i = 150$

$\Rightarrow \sum_{i=1}^{9} x_i + 25 = 150$

$\Rightarrow \sum_{i=1}^{9} x_i = 125$

$\Rightarrow \sum_{i=1}^{9} x_i + 15 = 140$

$$\text{Actual mean} = \frac{140}{10} = 14 = \bar{x}_{new}$$

$$\sum_{i=1}^{9} \frac{x_i^2 + 25^2 - 15^2}{10} = 15$$

$$\Rightarrow \sum_{i=1}^{9} x_i^2 + 625 = 2400$$

$$\sum_{i=1}^{9} x_i^2 = 1775$$

$$\sum_{i=1}^{9} x_i^2 + 15^2 = 2000 = \left(\Sigma x_i^2\right)_{actual}$$

$$6^2_{actual} = \frac{\left(\Sigma x_i^2\right)_{actual} - \left(\bar{x}_{new}\right)^2}{10}$$

$$= \frac{2000}{10} - 14^2$$

$$= 200 - 196 = 4$$

$$(S.D)_{actual} = 6 = 2$$

70. Let $x_1, x_2, x_3, ..., x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal $\bar{x}$ to is _____. **[JEE(Main)July2022]**

माना $x_1 = 3, x_2, x_3, ..., x_{20}$ एक गुणोत्तर श्रेढ़ी में हैं, जिसका सार्व अनुपात $\frac{1}{2}$ है। प्रत्येक x_i की जगह $(x_i - i)^2$. लेकर नये आँकड़े बनाए जाते हैं। यदि नये आँकड़े का माध्य $\bar{x}$ है तो महत्तम पूर्णांक $\leq \bar{x}$ है_____ **[JEE(Main)July2022]**

Ans. 142

Sol. $$\Sigma x_0^1 = \frac{3\left(1-\left(\frac{1}{2}\right)\right)^{20}}{1\frac{-1}{2}} = 6\left(1 - \frac{1}{2^{20}}\right)$$

$$= \sum_{i=1}^{20} \left(x_{i=1}\right)^2$$

$$= \sum_{i=1}^{20} \left(x_i\right)^2 + (i) - 2x_i i$$

$$\text{Now} = \sum_{i=1}^{20}(x_i)^2 = \frac{9\left(1-\left(\frac{1}{4}\right)\right)^{20}}{1-\frac{1}{4}} = 12\left(1-\frac{1}{2^{40}}\right)$$

$$\sum_{i=1}^{20} i^2 = \frac{1}{6}\times 20\times 21 + 41 = 2870$$

$$\sum_{i=1}^{20} x_i i = s = 3 + 2.3\frac{1}{2} + 3.3\frac{1}{2^2} + 4.3\frac{1}{2^3} + \ldots\ldots\text{AGP}$$

$$= 6\left(-\frac{22}{2^{20}}\right)$$

$$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12\left(2 - \frac{22}{2^{20}}\right)}{20}$$

$$\bar{x} = \frac{2858}{20} + \left(\frac{-12}{2^{40}} + \frac{22}{2^{20}}\right)\times\frac{1}{20}$$

$$\left[\bar{x}\right] = 142$$

71. Let the mean and the variance of 20 observations $x_1, x_2, \ldots x_{20}$ be 15 and 9, respectively. For $\alpha \in R$, if the mean of $(x_1+\alpha)^2, (x_2+\alpha)^2, \ldots\ldots, (x_{20}+\alpha)^2$ is 178, then the square of the maximum value of α is equal to _____.

[JEE(Main)July2022]

माना 20 प्रेक्षणों $x_1, x_2, \ldots x_{20}$ के माध्य तथा प्रसरण 15 तथा 9 हैं। $\alpha \in R$ के लिए, यदि $(x_1+\alpha)^2, (x_2+\alpha)^2, \ldots\ldots, (x_{20}+\alpha)^2$ का माध्य 178 है, तो α के अधिकतम मान का वर्ग बराबर है _____ **[JEE(Main)July2022]**

Ans. 4

Sol. $\sum x_1 = 15\times 20 = 300$ .....(i)

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \qquad \text{.....(ii)}$$

$$\sum x_1^2 = 234\times 20 = 4680$$

$$\frac{\sum(x_1+\alpha)^2}{20} = 178 \Rightarrow \sum(x_1+\alpha)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2\alpha\sum x_1 + \sum\alpha^2 = 3560$$

$$4680 + 600\alpha + 20\alpha^2 = 3560$$

$\Rightarrow \alpha^2 + 30\alpha + 56 = 0$

$\Rightarrow (\alpha + 28)(\alpha + 2) = 0$

$\alpha = -2, -28$

Square of maximum value of α is 4.

72. The sum and product of the mean and variance of a binomial distribution are 82.5 and 1350 respectively. Then the number of trials in the binomial distributions is _____. **[JEE(Main)July2022]**

एक द्विपद बंटन के माध्य तथा प्रसरण का योग तथा गुणनफल क्रमश: 82.5 तथा 1350 हैं। तो द्विपद बंटन में परीक्षणों की संख्या है _____ **[JEE(Main)July2022]**

Ans. 96

Sol. Let, mean = m = np

& variance = v = npq, p + q = 1

$\text{Sum} = m + v = \frac{165}{2}$

Product = mv = 1350

On solving,

$m = np = 60$ & $v = npq = \frac{45}{2} \therefore q = \frac{3}{8} \therefore P = \frac{5}{8}$

Hence n = 96